

L3



CLASS X - SCIENCE

CONTROL AND COORDINATION

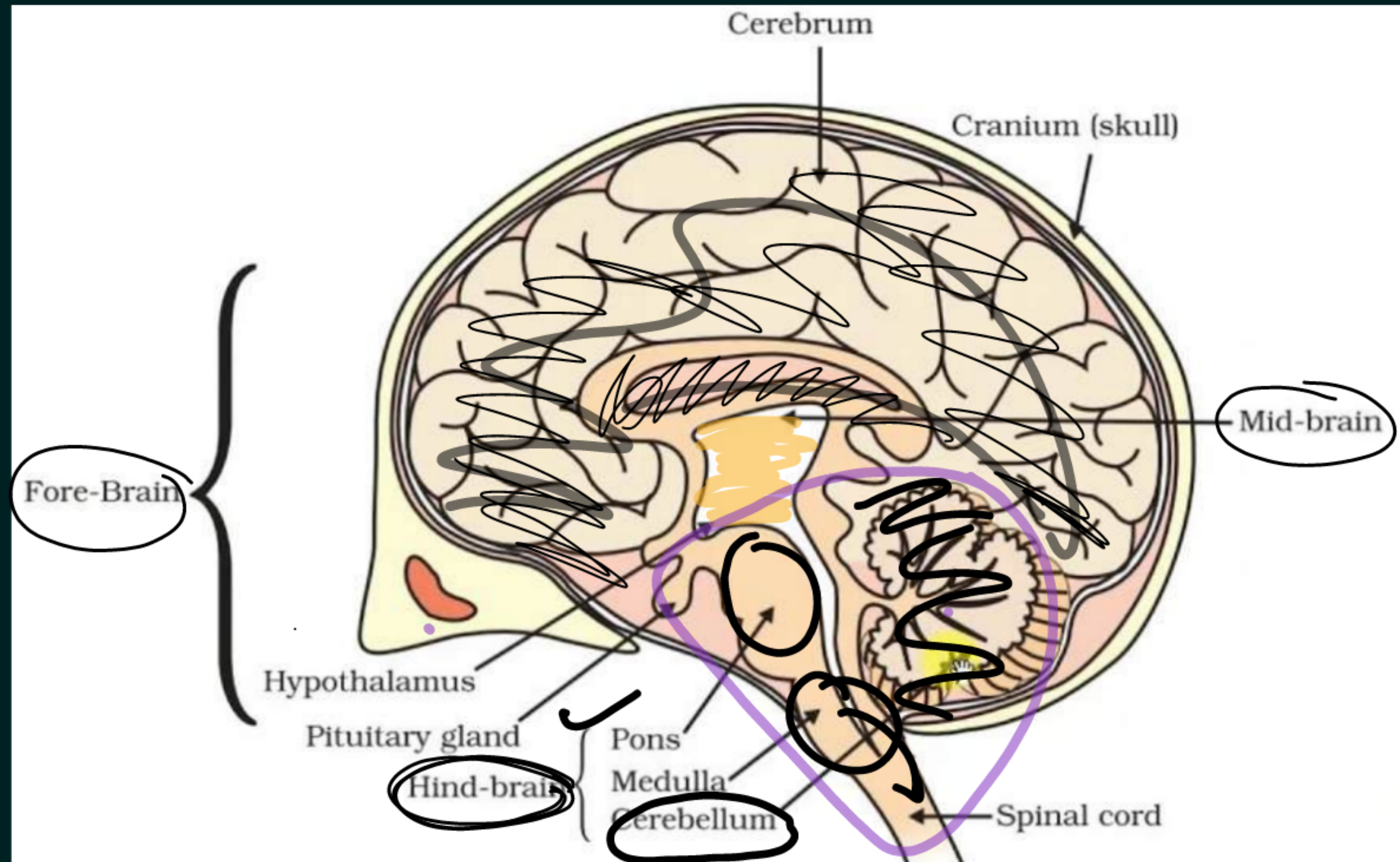
| PRASHANT KIRAD |

Brain

- **Main control center** of the nervous system
- **Receives and processes information** from the body
- **Controls voluntary and involuntary actions**
- Maintains **balance and coordination**
- Responsible for **thinking, memory, learning, and emotions**
- Interprets signals from **sense organs**
- Regulates **heartbeat, breathing, and other vital activities**



Human brain structure



Major parts of Brain

Fore-brain

Mid-brain

Hind-brain

- Cerebrum
 - Hypothalamus
 - Thalamus
- 90%*
- Hunger Center*

- Cerebellum
- Pons
- Medulla

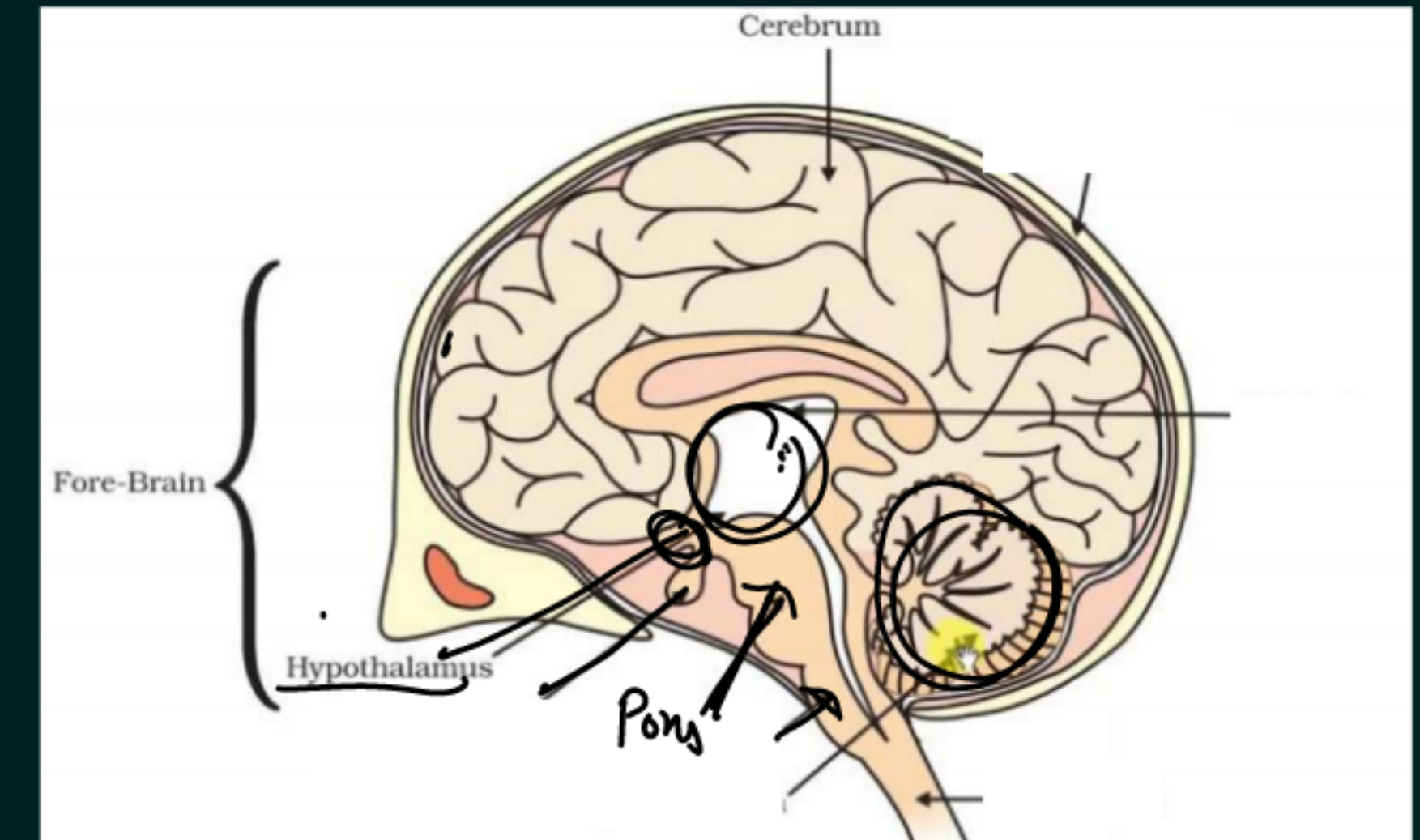
Corpus Callosum

- The corpus callosum is a thick band of nerve fibres in the brain.
- It connects the left and right hemispheres of the cerebrum.
- Its main function is to allow communication between the two halves of the brain, so that they can work together



Fore brain

- It is the main thinking part of the brain.
- Has regions that receive sensory impulses from various receptors.
- Contains separate specialised areas for hearing, smell, sight, etc.
- Has association areas where sensory information is interpreted by combining it with:
 - information from other receptors, and
 - information already stored in the brain.
- Makes decisions about how to respond.
- Passes information to the motor areas, which control the movement of voluntary muscles (e.g., leg muscles).
- Contains a separate hunger centre that gives the sensation of feeling full.



Midbrain

The midbrain is the middle part of the brain, located between the forebrain and hindbrain.

Functions of the Midbrain:

- ✓ Connects Fore and Hind Brain
- Controls many involuntary actions (along with the hind-brain). → Blinking
- Involved in controlling reflex movements of the head, neck, and eyes (e.g., reflex actions of the eye).



Hind brain

The hindbrain is the lower part of the brain, located at the back of the skull. It connects the brain to the spinal cord.

Controls involuntary actions through its parts:

Pons →
✓ Medulla: oblongata

1. Controls involuntary actions such as blood pressure, salivation, and vomiting.

→ Balance

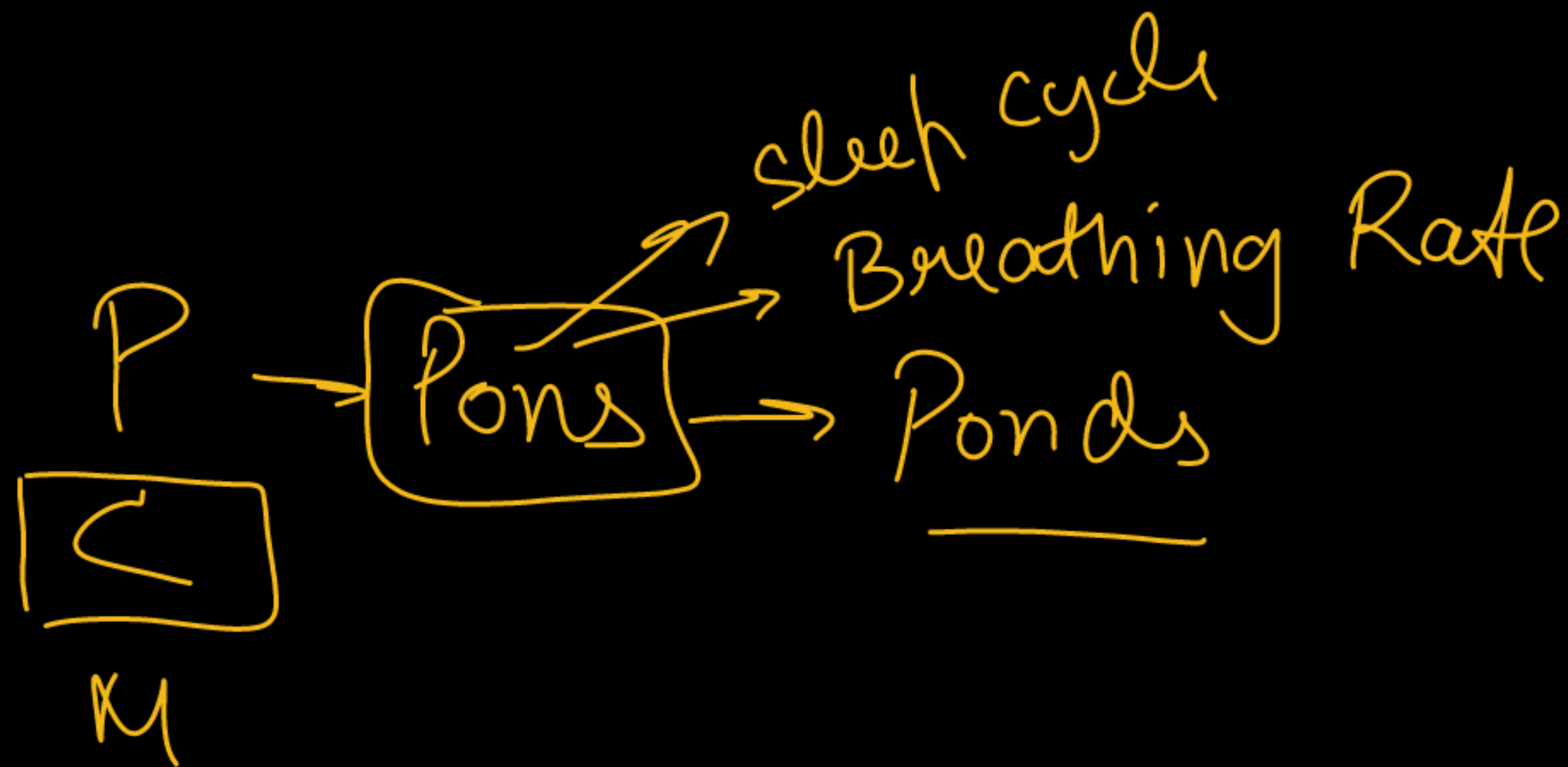
Cerebellum

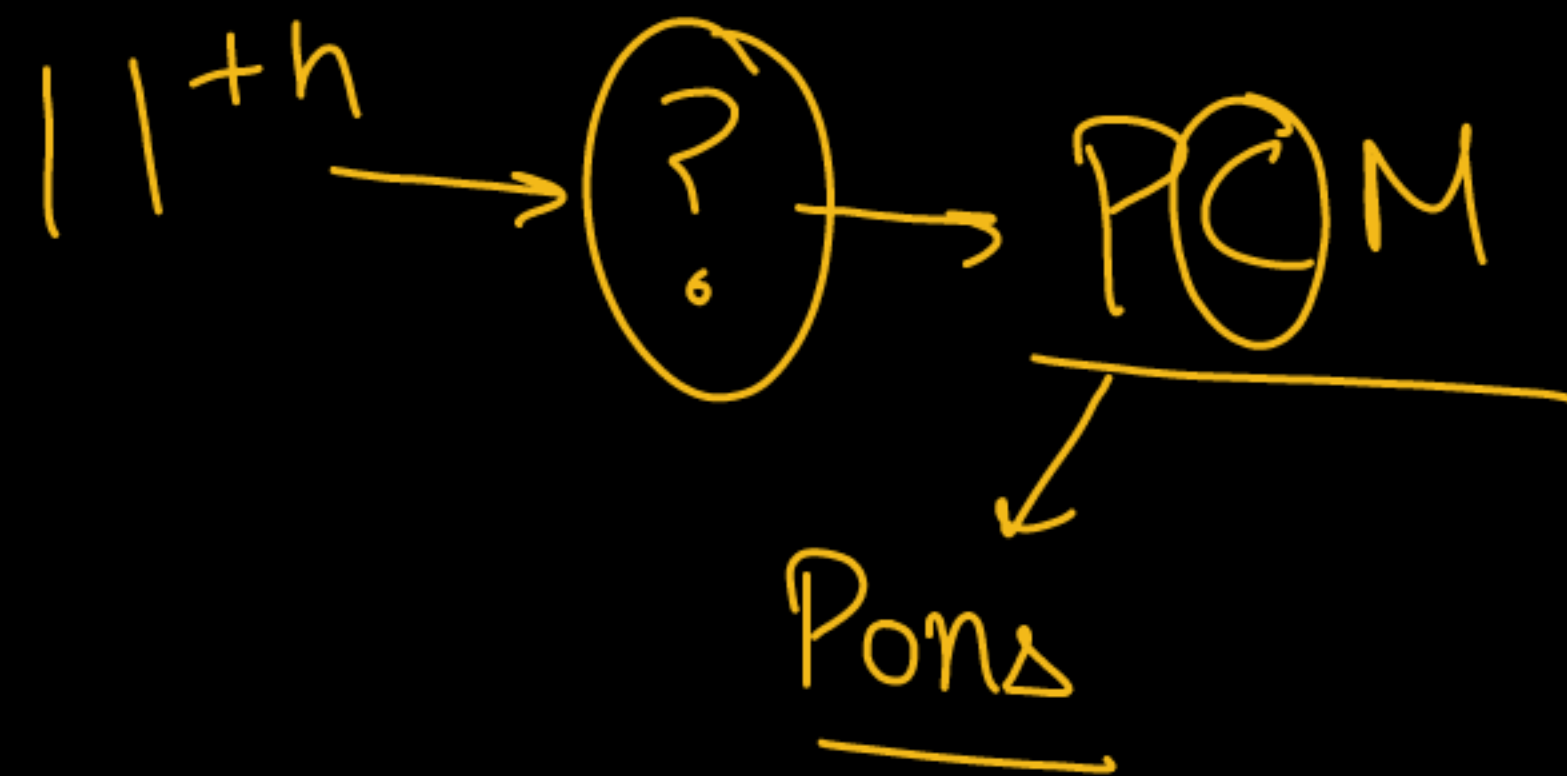
1. Responsible for precision of voluntary actions (e.g., picking up a pencil, writing).

2. Maintains posture and balance of the body (e.g., walking in a straight line, riding a bicycle).

Pe (N)







Spinal cord



Spinal cord

- The spinal cord is a cylindrical extension of the brain that begins from the medulla oblongata.
- It is a part of the central nervous system and is protected by the vertebral column and meninges.
- A total of 31 pairs of spinal nerves arise from it.
- It acts as a pathway for transmission of nerve impulses between the brain and body parts.
- It controls spinal reflexes through reflex arcs.
- It helps the body respond quickly to stimuli.



Protection of nervous organs

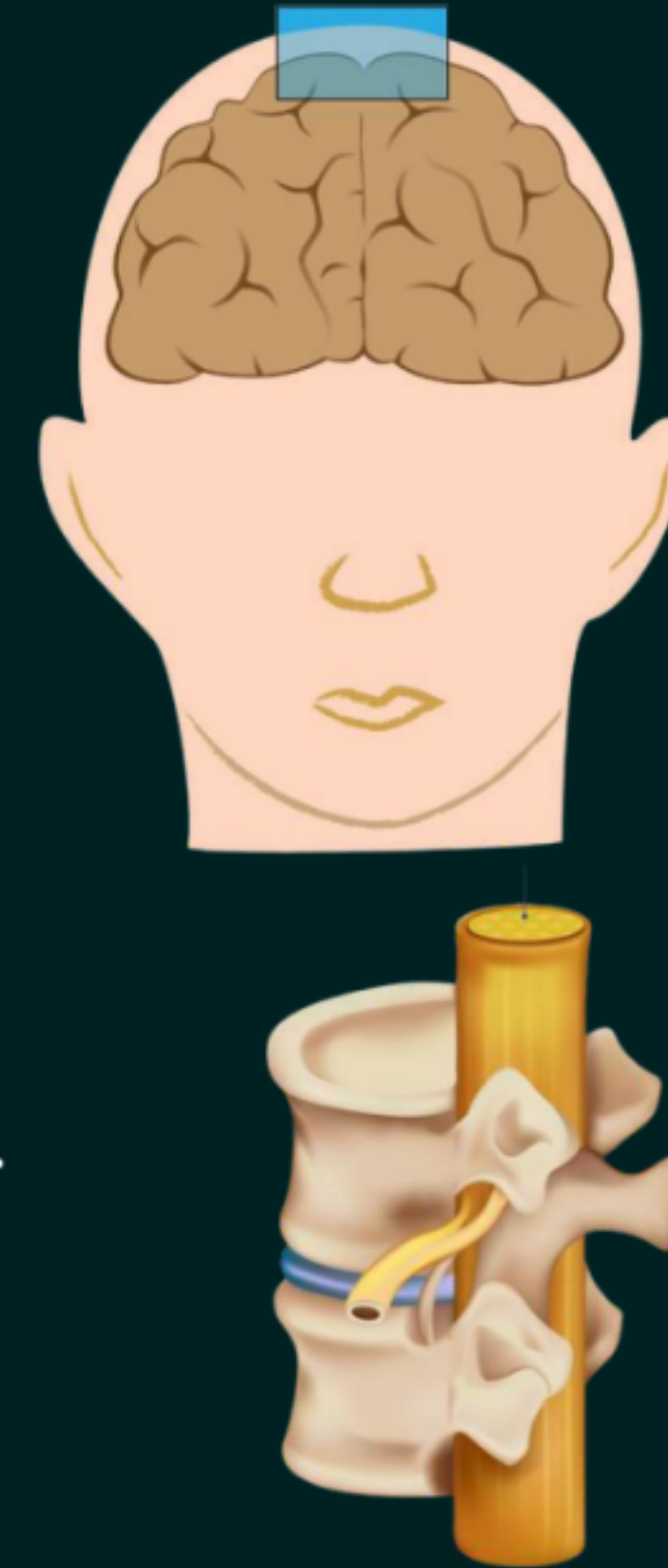
Q: How are delicate tissues like the brain and spinal cord protected?

Protection of the Brain:

- The brain is protected by the bony skull (cranium).
- It is covered by three membranes (meninges).
- A fluid called cerebrospinal fluid (CSF) surrounds it, which acts as a shock absorber and protects it from injury.

Protection of Spinal Cord

- The spinal cord is protected by the vertebral column (backbone).
- It is also covered by meninges.
- Cerebrospinal fluid (CSF) around the spinal cord provides cushioning and prevents damage.



LET'S TEST YOUR MEMORY



Q. Which parts of the brain control blood pressure?

A → Spinal cord, skull, hypothalamus

~~C~~ → Pons, medulla, cerebellum

B → Cord, skull, cerebrum

D → Pons, medulla, pituitary

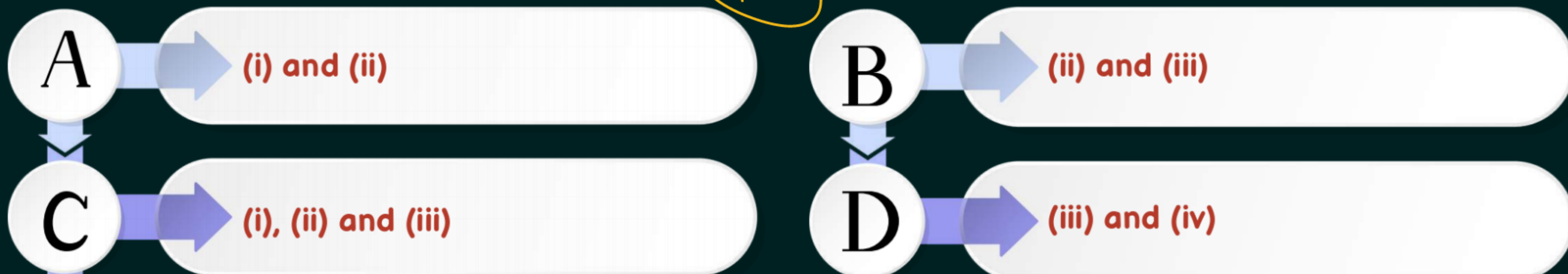
LET'S TEST YOUR MEMORY



Q. Which of the following statements are true about the brain?

- ☒ (i) The main thinking part of brain is hind brain
- ☒ (ii) Centres of hearing, smell, memory, sight etc are located in fore brain.
- ☒ (iii) Involuntary actions like salivation, vomiting, blood pressure are controlled by the medulla in the hind brain
- ☒ (iv) Cerebellum does not control posture and balance of the body

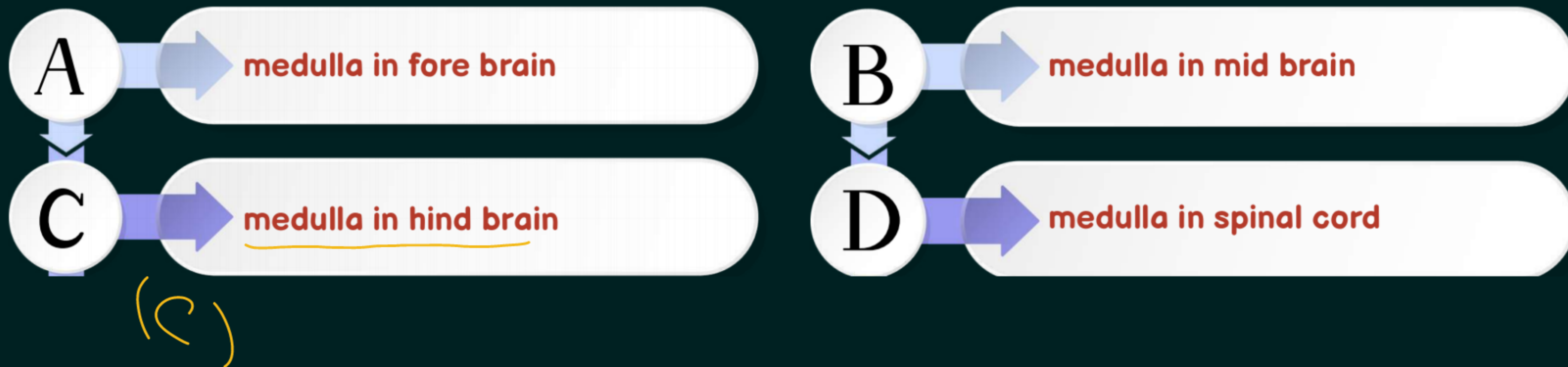
B



LET'S TEST YOUR MEMORY



Q. Involuntary actions in the body are controlled by



LET'S TEST YOUR MEMORY

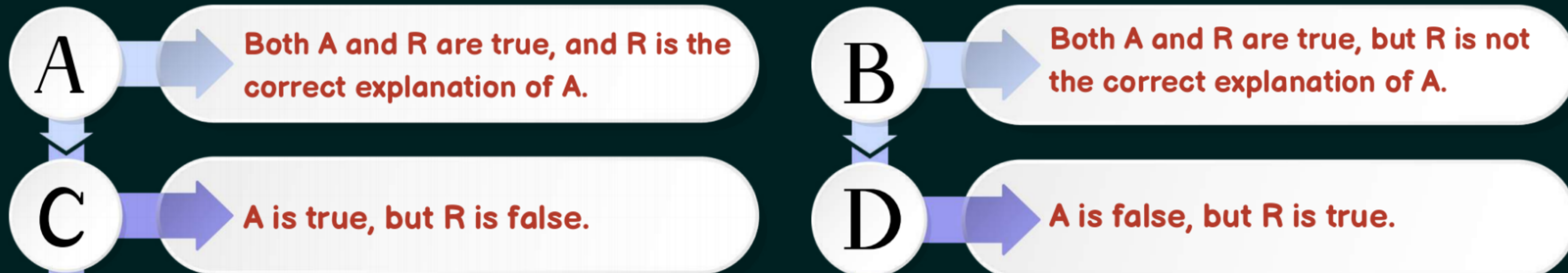


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Q. Assertion (A): The brain is the main coordinating center of the body.

Reason (R): The spinal cord sends signals from brain to the rest of the body. [CBSE 2022]

B

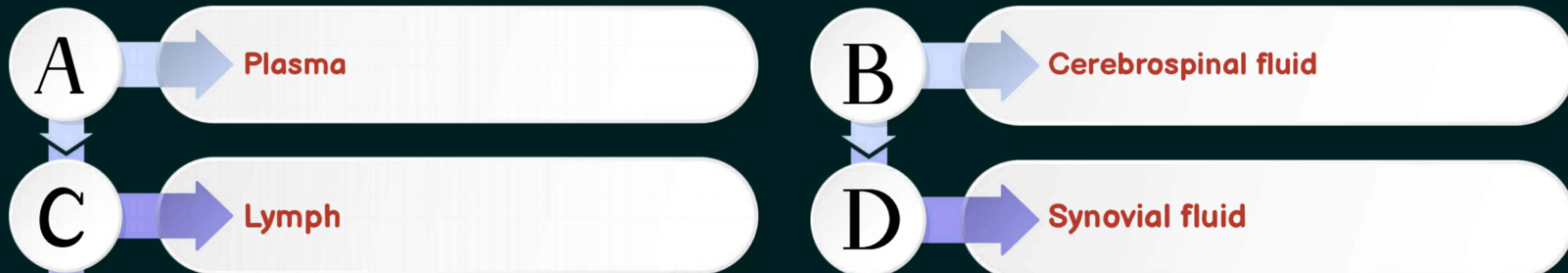


LET'S TEST YOUR MEMORY



Q. Which fluid acts as a cushion and protects the spinal cord from shocks?

B



LET'S TEST YOUR MEMORY



Q. Which part of the human brain is responsible for: [CBSE 2025]

- (a) Thinking → Fore →
- (b) Picking up a pencil
- (c) Controlling blood pressure
- (d) Controlling hunger
- (e) Vomiting

Function	Part of Brain Responsible
(a) Thinking	Cerebrum / Forebrain
(b) Picking up a pencil	Cerebrum / Forebrain
(c) Controlling blood pressure	Medulla
(d) Controlling hunger	Hypothalamus
(e) Vomiting	Medulla

LET'S TEST YOUR MEMORY



Q. A person is making a list to purchase a few things from a nearby market. Explain how the forebrain plays an important role in performing this activity. (CBSE 2025)

Ans:

The forebrain is the main thinking part of the brain. It receives sensory information, interprets it, thinks about needs, and helps in decision-making. It also controls reasoning, memory, and voluntary actions such as writing a list.

LET'S TEST YOUR MEMORY



Q. (a) How is the brain protected in our body?

(b) A doctor finds in one of his patients that he is not maintaining a proper posture and balance of his body. State the region of brain and also the part of brain which is responsible for it.

(CBSE 2025)

Ans: (a) Protection of the Brain

- The brain is protected by the skull and cerebrospinal fluid (CSF).
- The skull forms a hard protective covering around the brain.
- CSF surrounds the brain and acts as a shock absorber.

(b) The patient is unable to maintain posture and balance.

- This function is controlled by the hindbrain.
- The part responsible is the cerebellum.

LET'S TEST YOUR MEMORY



Q. What will happen if the cerebellum is damaged? State any two effects.

Ans. If the cerebellum is damaged:

- The person will lose balance and posture.
- Voluntary actions will lose precision and coordination.

LET'S TEST YOUR MEMORY



Q. State one main function of: (CBSE 2023)

1. Medulla →
2. Cerebellum

Part	One Main Function
Medulla	Controls involuntary actions like heartbeat and breathing
Cerebellum	Maintains posture and balance of the body



LET'S TEST YOUR MEMORY

HW

Q. Case Study -

A 10-year-old child tries to ride a bicycle for the first time. Initially, he struggles to maintain balance, but over time, he improves his coordination and does not fall. This learning process is due to the role of a specific part of the brain responsible for posture, balance, and muscle coordination.

1. Which part of the brain helps the child maintain balance?

- a) Cerebrum
- b) Medulla
- c) Cerebellum
- d) Hypothalamus

2. What type of action is involved in learning to ride a bicycle?

- a) Involuntary action
- b) Reflex action
- c) Voluntary action
- d) Hormonal action

3. The cerebellum is a part of which major division of the brain?

- a) Forebrain
- b) Midbrain
- c) Hindbrain
- d) None of the above

LET'S TEST YOUR MEMORY



Q.5

Q. The image given below represents a section of the human brain, the highest coordinating centre in the human body.

(A) Which part of the human brain includes the most complex and specialized part of the whole system?

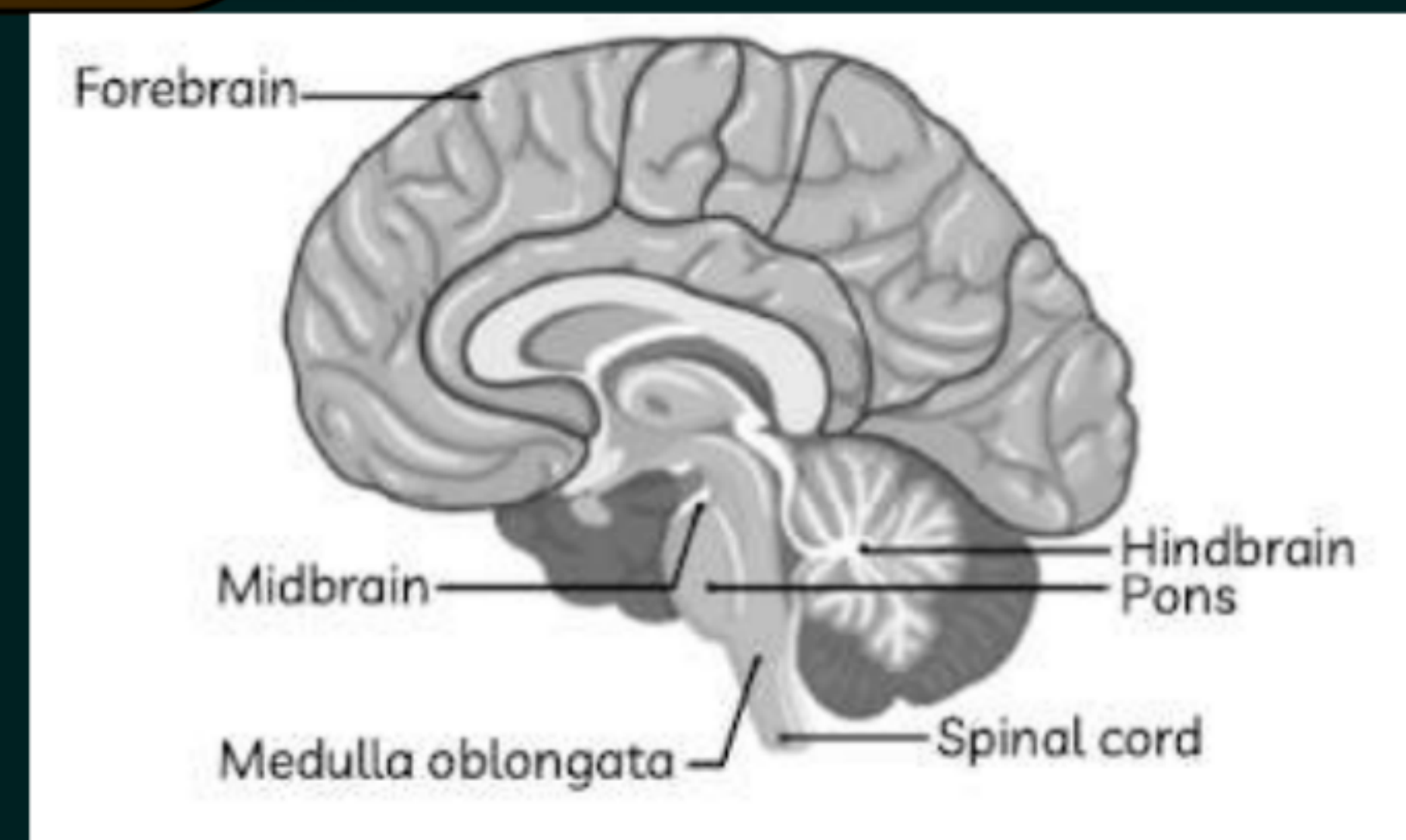
(B) Which part is responsible for the sensation of feeling full?

(C) Mention two functions of the midbrain.

(D) Several regulatory centres for a variety of life processes are present in the different sections of the human brain. Which part contains the regulatory centre responsible for:

(i) Sneezing and vomiting?

(ii) Adjustment of movement and posture?



(A) The Forebrain (Cerebrum) includes the most complex and specialized part of the whole system.

(B) The Hypothalamus (part of the forebrain) is responsible for the sensation of feeling full.

(C) Two functions of the Midbrain:

1. Controls reflex movements of the head, neck and trunk in response to visual and auditory stimuli.
2. Acts as a relay centre for visual and auditory information.

(D) (i) Medulla oblongata - Sneezing and vomiting.

(ii) Cerebellum (Hindbrain) - Adjustment of movement and posture.

Actual image of the inside of my brain

